

CSA1014 SOFTWARE ENGINEERING

CO-1 ASSESSMENT TOOL - 2

Warehouse Robotics Coordination and Inventory Automation Platform

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Project Vision & Objectives

Product Vision Statement

To deliver an intelligent, highly resilient multi-robot coordination system that desynchronizes latency and prevents traffic gridlocks in automated fulfillment warehouses.

Ensures seamless continuous multi-brand AMR fleet operations and accurate real-time inventory tracking.

Strategic Problem Alignment

Fulfillment Bottlenecks: Addresses manual log delays and fleet stalls at narrow junctions.

Core Objective: Achieve sub-100ms fleet path calculation and 99.9% inventory sync accuracy under high pick-volume conditions.

Stakeholder Identification

Stakeholder Role	Primary Goal / Need	System Interaction Focus	Impact Level
Warehouse Floor Manager	Maximize daily order pick throughput	Live fleet telemetry & bottleneck alerts	High (Operational)
Robotics Systems Engineer	Maintain zero-collision fleet routing	ROS2 node monitoring & emergency stops	Critical (Technical)
Inventory Operator	Eliminate lost stock desynchronization	Automated RFID tag updates & bin logs	High (Data Quality)
Product Owner	Deliver incremental Agile value per sprint	Backlog prioritization & sprint goals	Executive (Strategy)

Core System Epics



Epic 1: Multi-AMR Pathing

Centralized dynamic grid routing, conflict detection, and intersection deadlock prevention algorithms for active robot fleets.



Epic 2: Inventory Sync

Real-time RFID and vision-assisted inventory updates synced directly with Enterprise Resource Planning (ERP) databases.



Epic 3: Telemetry & E-Stop

Safety override controls, latency drop recovery, and real-time operational dashboard for floor managers.

Agile User Stories

Functional User Stories

US-01 (Pathfinding): As a Robotics Engineer, I want dynamic rerouting at intersections so that AMRs do not collide or lock up aisles.

US-02 (Tracking): As an Inventory Operator, I want shelf pick scans to stream instantly so stock counts remain accurate.

US-03 (Overrides): As a Floor Manager, I want a single-click emergency pause so I can safely enter active robot zones.



Acceptance Criteria Definition



US-01: Collision Prevention (Pathfinding)

Given two AMRs approach an intersection within 2 meters, *When* route conflict is detected, *Then* lower priority AMR stops within 200ms and recalculated path is assigned.



US-02: Instant Inventory Log (Sync)

Given an AMR picks a bin payload, *When* RFID scanner registers the item tag, *Then* database updates item status to 'In-Transit' within <100ms.



US-03: Zone Telemetry Emergency Stop

Given an emergency manual override is triggered on UI, *When* broadcast packet reaches fleet, *Then* all robots in target zone halt within 500ms.

Estimation & Sprint Goal

31 SP

Sprint 1 Target Velocity

Sprint Goal & Execution Plan

Sprint Goal: Demonstrate a zero-collision 10-robot simulated grid navigation with real-time inventory updates on dashboard.

Estimation Methodology: Planning Poker using modified Fibonacci sequence (1, 2, 3, 5, 8, 13) based on technical complexity and risk.

Deliverables: Functional ROS2 simulation build, verified Given-When-Then test suite, and telemetry API docs.

Jira Implementation

Jira

Starred

Apps

Plans

Spaces

Recent

Warehouse basic

warehouse robotics coordinatio...

scrum

kanban

kanban 2

More spaces

Recommended

Collect IT requests

Filters

Dashboards

Teams

Goals

Projects

Search

+ Create

Choose a plan

JP

Spaces

Warehouse basic

Summary

List

Board

Backlog

Development

Forms

More 3

Search backlog

JP

Filter

WB Sprint 1

Add dates

(5 work items)

0

0

0

Start sprint

WB-4

dashboard & reporting

To Do

-

WB-3

Robot Coordination

In Progress

-

WB-2

Task scheduling

Done

-

JP

WB-5

automation of works

In Progress

-

Pri

WB-12

inventory management

To Do

-

+ Create

Backlog

(1 work item)

0

0

0

Create sprint

Jira work item

Add epic /

1

WB-5

automation of works

In Progress

Improve Task

Description

Add a description...

Attachments

Add attachment

THANK YOU